

MUHAMMAD AQIL SYARIMAN

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SUMMARY

Undergraduate AI/GenAI Engineer with demonstrated experience building production RAG pipelines, LLM-powered applications, and vector search systems using LangChain, ChromaDB, LLaMA, and Groq API. Currently interning in last semester as a Software Engineer. Demonstrated ability to design and deploy end-to-end AI systems from data ingestion to cloud deployment. Proficient in Python, C, automation, full-stack development and Agile.

EDUCATION

Universiti Malaysia Perlis (UniMAP)

Bachelor Of Electronic Engineering Technology | (2022 - Present)

- Expected Graduation: Nov 2026 | CGPA: 3.63
- Related Coursework: Artificial Intelligence, Object-Oriented Programming, Embedded Software Technology, Computer Architecture, Data Structure, Operating Systems, Cloud Infrastructure and Services.

Diploma in Computer Engineering | (Graduated)

PROFESSIONAL EXPERIENCE

Centific Global Solutions, Inc. (Software Engineer Intern)

(February 2026 - August 2026)

- Engineered PowerShell automation scripts for Microsoft SCCM, leveraging AI tools (GitHub Copilot) to accelerate code development and successfully **reducing** manual execution time by **50%**.
- Executed and managed **1,000+** test cases, utilizing Microsoft SQL Server queries to validate system results and ensure data integrity.
- Provided critical infrastructure support by assisting in SCCM deployment, system validation, and advanced troubleshooting.

CTRM Aero Composites Sdn Bhd (Software Intern)

(January 2022 - April 2022)

- Optimized front-end architecture using React and JavaScript for an aerospace manufacturing firm
- Executed end-to-end API and database validation using Postman and SQL queries

ACADEMIC PROJECTS

Semiconductor AI Intelligence Platform | Python, LangChain, ChromaDB, FlashRank, LLaMA 3.3 70B, Groq API, Streamlit

[Live Demo](#) | [GitHub](#)

- Built a multi-document RAG pipeline to ingest Intel and Micron semiconductor datasheets, implementing function-level chunking with page and source metadata tracking
- Integrated FlashRank re-ranking pipeline to improve retrieval accuracy across overlapping technical documents
- Engineered a semiconductor expert prompt system that returns precise answers with exact document and page citation
- Deployed on Streamlit Cloud with persistent chat memory and multi-PDF support

RAG Assistant | Python, LangChain, ChromaDB, Sentence Transformers, PyPDF2, Groq API (LLaMA 3.1)

[Live Demo](#) | [GitHub](#)

- Built a foundational RAG pipeline to ingest PDF documents, implementing automated text extraction, 500-character chunking, and high-dimensional vector embedding generation
- Designed semantic search system using ChromaDB with cosine distance metrics for context-aware document retrieval
- Implemented hallucination prevention by filtering out-of-boundary queries before LLM inference
- Deployed on Streamlit Cloud with PDF upload support and interactive chat interface

Smart Poultry Habitat | C, ESP32, React Native, IoT

[Github](#)

- Led a team programming ESP32 microcontroller in C for real-time environmental sensor data processing
- Developed bidirectional communication pipeline between embedded hardware and React Native application

Student Mental Well-Being Monitoring System with Analytics | React Native, TypeScript, Firebase, REST APIs

[Github](#)

- Directed end-to-end Agile development of a React Native application implementing modular component architecture and continuous integration testing
- Designed real-time data synchronization pipeline to stream biometric telemetry from wearable devices to Firebase Firestore via REST APIs
- Engineered analytics dashboard to aggregate and visualize historical biometric trends using reactive data-binding UI components

AI-Based Loan Approval System | Python, Scikit-learn, Pandas, NumPy

- Built predictive ML pipeline to preprocess imbalanced financial datasets and train classification models for loan default risk evaluation

AI-Powered Construction Safety Monitoring System | AWS Rekognition, S3, Lambda, Python

- Architected cloud-native computer vision solution for PPE detection using AWS Rekognition, S3, and Lambda.

TECHNICAL SKILLS

Languages:

- C, Python, PowerShell, React, JavaScript/TypeScript, Node.js, SQL.

Hardware & Embedded:

- Hardware & Embedded: ESP32, Arduino, Microcontroller Programming

AI/GenAI:

- LangChain, ChromaDB, Groq API, LLaMA 3.1/3.3, Sentence Transformers, FlashRank, RAG Pipelines, Prompt Engineering, Vector Embeddings

Software Tools:

- Git, GitHub, Postman, Microsoft SSMS, AzureDevOps

Cloud & Networking:

- AWS (S3, Lambda, Rekognition), GCP, Firebase, Cisco Routing, OSPF

AWARDS / CO-CURRICULUM

- Dean's List for 5 out of 6 semesters
- Silver Award Winner (Leader) – IoT Design Project
- Facilitator at Universiti Malaysia Perlis programs

CERTIFICATES

- The Complete Full-Stack Web Development Bootcamp
 - Dr. Angela Yu | Udemy | Nov 2025